Assignment 2 - Advanced Machine Learning 2026

Instructions

- In this assignment, you will develop and compare machine-learning approaches for Parkinson's Disease (PD) classification, including preprocessing, model development, validation and hyperparameter optimization, threshold selection, evaluation, and interpretation.
- Assignments are to be done in groups of 5.
- Use Python to solve the exercise.
- Deliverables:
    - The file **Team-XX.pdf** (where XX is your group number, e.g. Team-07.pdf) containing your report.
    - A file **Team-XX-text.pdf** containing the same written content as your report, excluding tables and figures. This will be used to check whether you adhere to the **7-page maximum**.
    - Python code that you have developed to solve the assignment. Keep the code of methods that did not make it into the final model, but do not submit code that does not work. Make sure that this code is readable and concise. If you developed multiple files, place them in a zip file named **Team-XX-code.zip**.
    - Submit all deliverables electronically through Canvas. Make sure to submit everything in a single zip file named **Team-XX-A2.zip**.
- Deadline for submission of solutions: **Sunday, December 6th, 2026, 20:00 hrs.**
- Note that if you have performed some parts of the assignment but do not discuss this in your report, you will not receive the points associated. We will only evaluate your report; the code will only be consulted for verification purposes.
- Be concise and to the point.
- Use correct terminology and refer to scientific literature where appropriate.

Questions regarding the assignment can be asked through Canvas or during the sessions on Thursday mornings.

Student Assignment 2

You are working as a data scientist on a healthcare analytics project. Your task is to develop and compare machine-learning approaches for predicting Parkinson's Disease (PD) status from structured clinical variables. You will investigate SVM, MLP, and a Fuzzy Inference System (FIS), compare their performance, and determine appropriate decision thresholds. Your focus should be on applying a sound machine-learning workflow, evaluating the models appropriately, and explaining how their outputs could support a hypothetical decision-making process. Because the dataset is synthetic, the resulting models should not be interpreted as clinically validated diagnostic tools.

The dataset

- You will use the synthetic Parkinson's Disease dataset provided on Canvas.
- Each row represents one synthetic patient record containing demographic, lifestyle, clinical, cognitive, functional, and symptom-related variables.
- The prediction target is Diagnosis, where 0 = Non-PD and 1 = PD.
- PatientID is an identifier and DoctorInCharge is a constant/confidential field; neither should be used as a predictive feature.
- A separate .txt file provided on Canvas contains descriptions of all dataset attributes.

1) Problem framing & EDA

- Define the prediction target (PD vs. Non-PD) and describe an appropriate intended use for the predictive system. Explain how the model would fit into the workflow you propose, discuss the possible consequences of false-positive and false-negative predictions, and justify which type of error is most important in that context.
- Choose appropriate evaluation metrics and justify your choices.
- Exploratory Data Analysis (EDA): uncover patterns, spot anomalies such as outliers, and identify relationships between variables using concise tables and plots.

2) Preprocessing & feature engineering

- Select an appropriate preprocessing pipeline and justify your choices.

Tip: For SVM and MLP, use a scikit-learn pipeline so that preprocessing is fitted using the training data within each validation fold, helping to avoid data leakage. If different groups of features require different preprocessing steps, you may use ColumnTransformer to apply the appropriate transformation to each group of columns. For the FIS, determine any data-dependent preprocessing (e.g., scaling or ranges used to define membership functions) using the training data only, and then apply those settings unchanged to the validation and test data.

3) Models

- SVM (SVC): train and tune both linear and RBF kernels. Treat them as candidate configurations within the SVM model family and select the best SVM configuration for the final comparison with MLP and FIS.
- MLPClassifier.
- Fuzzy Inference System (FIS) using 4-6 continuous inputs whose relevance to Parkinson's Disease can be reasonably motivated using the provided data description and/or relevant

literature. Define 2-3 membership functions per input and 4-8 rules; defuzzify via centroid and map the output to [0,1] as a risk score.

Note: For the FIS, justify the selected inputs and rule design using an appropriate source, such as the provided data description and/or relevant literature. You are **not expected to establish clinical validity or consult clinical experts**. The main emphasis is on **source-to-model traceability**: clearly explain how the information from your source is translated into the selected inputs, membership functions, and fuzzy rules.

4) Validation & Hyperparameter Optimization (HPO)

- Reserve a held-out test set before model tuning and selection. Use only the remaining training/validation data for model development; do not use the test set for hyperparameter tuning or model selection.
- For SVM and MLP, use stratified cross-validation and a limited hyperparameter search. Evaluate no more than 20 hyperparameter configurations for the SVM model family in total and no more than 20 configurations for MLP. Motivate your validation strategy and hyperparameter-search choices.
- For the FIS, evaluate and refine the model using only the training/validation data, and clearly explain the validation procedure you use. Do not modify the FIS after inspecting the test-set results.

5) Model selection & thresholding

- After selecting the best configuration for each model, obtain validation predictions or scores using only the non-test data. You may use a held-out validation set or out-of-fold predictions from the training data.
- For comparability, all groups must use the same threshold-selection rule: for each model, select the threshold that maximizes F_2 on the validation predictions. The rule is the same for everyone, but the resulting numerical threshold may differ across models and groups.
- Once selected, freeze each model's threshold. Do not change it after inspecting the test set.
- Evaluate the final SVM, MLP, and FIS models on the test set using their respective frozen thresholds. Report the confusion matrix, recall, precision, specificity, and F_2.

6) Decision support & practical value

- Based on your final comparison, identify the model you would recommend for the workflow defined in Task 1 and justify your choice using the model results.
- Identify who would use or be affected by the model output and describe the decision or action that the output could support.
- Propose and justify a simple decision policy that connects the model output to that action.
- Discuss the consequences or cost implications of false-positive and false-negative predictions in your chosen workflow.

- Explain how your model results and chosen decision policy could create value and state the assumptions underlying your argument.

7) Reflection

- What worked best, and why? Discuss factors that may have influenced model performance, such as bias-variance trade-offs, linearity, model capacity, regularization, and the amount of training data.
- What challenges did you encounter while carrying out the assignment, and how did you address them?
- What are the main limitations of your models, and what future improvements would you propose?

Report Guidelines

Prepare a report in which you explain the modeling process, results, and applicability of your approach. Show in your report that you have performed and discussed all of the tasks mentioned above.

These guidelines apply to both submitted report versions. **Team-XX.pdf** is the complete report, while **Team-XX-text.pdf** must contain the same written content and section structure but with all tables and figures removed.

- The written content of the report has a maximum length of **7 A4 pages**, excluding tables and figures. Compliance with this limit will be checked using **Team-XX-text.pdf**.
- Include only tables and figures that are necessary to support your analysis in **Team-XX.pdf**. Every table and figure must be clearly referenced and discussed in the main text.
- Include at least:

    1. **Problem framing & EDA**
    2. **Preprocessing & feature engineering**
    3. **Models**
    4. **Validation & hyperparameter optimization**
    5. **Model selection & thresholding**
    6. **Decision support & practical value**
    7. **Reflection**

These sections should be clearly represented in the report. Additional subsections may be included where appropriate. Make sure that the results and discussion from each task are presented in the relevant section.

You will be working with several modeling and validation choices throughout this assignment - **start early!**

Good luck!